

# Day 3 — The Finger Trick

**Module:** Multiply-Near-Ten (Primary) · **Time:** 30 min

## Goal

Children learn the hand-multiplication trick for any pair from 6 to 9.

## Materials

- Both hands (everyone has them).
- Class wall-poster: a hand diagram showing fingers folded down for  $7 \times 8$ .
- Worksheet: 16 problems (all combinations  $6 \times 6$  through  $9 \times 9$ ), with hand-diagram answer key.

## The flow

### Settle + chant (5 min)

The Twins chant + lightning round of 10 twins.

### Story (8 min) — “The Magic Hand Trick”

- “Today I’ll teach you a magic trick. You can multiply  $7 \times 8$  just by counting fingers. Watch.”
- **Demo:  $7 \times 8$ .**
  - “Hold up both hands. We’re going to fold down some fingers.”
  - “For 7: fold down  $(7 - 5) = 2$  fingers on the LEFT hand. So 2 down, 3 up on the left.”
  - “For 8: fold down  $(8 - 5) = 3$  fingers on the RIGHT hand. So 3 down, 2 up on the right.”
  - “Count the fingers DOWN on both hands together:  $2 + 3 = 5$ . That’s the TENS. → **50.**”
  - “Count the fingers UP on left  $\times$  fingers UP on right:  $3 \times 2 = 6$ . That’s the UNITS. → **6.**”
  - “Total:  $50 + 6 = 56$ . ✓”
- Repeat for  $6 \times 9$ :
  - 6: 1 down, 4 up. 9: 4 down, 1 up.
  - Down sum:  $1 + 4 = 5 \rightarrow 50$ .
  - Up product:  $4 \times 1 = 4$ .
  - Answer: **54.** ✓
- Repeat for  $8 \times 9$ :
  - 8: 3 down, 2 up. 9: 4 down, 1 up.
  - Down sum:  $3 + 4 = 7 \rightarrow 70$ .

- Up product:  $2 \times 1 = 2$ .
- Answer: **72. ✓**

### Practice (12 min)

- Whole class: teacher calls  $7 \times 6$ . Children put up hands and find the answer (42).
- 5 problems chorus:  $9 \times 9$  (81),  $6 \times 7$  (42),  $8 \times 8$  (64),  $9 \times 6$  (54),  $7 \times 9$  (63).
- Pair drill (last 7 min):
  - Partner A calls a problem (any from  $6 \times 6$  to  $9 \times 9$ ).
  - Partner B answers using the finger trick.
  - Swap.

### Show + chant (5 min)

- “Who got  $9 \times 9 = 81$  using their hands?” — hands up.
- Final chant.

### Homework

- Teach the finger trick to a parent or sibling. Multiply  $7 \times 8$  together.

### Differentiation

- **Tier up:** Try  $5 \times 6$  — what does the trick give? (5: 0 down, 5 up. 6: 1 down, 4 up. Down sum  $1 \rightarrow 10$ . Up product  $5 \times 4 = 20$ . Total 30. ✓) Notice: works for 5+ too. What about  $5 \times 5$ ? (Both 0 down. Down sum 0. Up product  $5 \times 5 = 25$ . ✓ Total 25.) Cool — works for 5 too.
- **Tier down:** Stay with  $7 \times 8$  and  $9 \times 9$  until comfortable. Don’t push variety today.

### Teacher’s quick notes

- A few children will have trouble keeping track of “down vs up.” Slow down with them; the trick *is* tricky on the first day.
- Some children will object: “I already know my 7 times table.” Honest response: “*Great! Now you have TWO ways of finding the answer.*”
- Don’t explain why it works today. Day 4 is the dot-drawing day.

### Sanskrit term mention

(Hold for Day 6.)